Appendix

Flexible regulation as a double-edged sword: The case of the organic seed standard

Survey sample

1. Survey sampling method

Survey data collection was spearheaded by Organic Seed Alliance (OSA), with a number of collaborators, using a mix of convenience and random sampling across the three time periods.

- 2010: Data were collected by convenience sample only, shared by 23 organizations, including non-profits and foundations, organic certifiers, and departments of agriculture. Surveys were distributed primarily over email, but paper, phone, and in-person surveys were conducted at the request of the grower. We received a total of 1,027 responses (712 web-based, 315 paper-based), 546 of which included complete enough data for our analysis.
- 2015: Data were collected by both random and convenience sampling. For the random sample, 800 certified organic producers were randomly selected from the approximately 12,000 organic producers that use seed, recorded in the NOP database. To encourage responsiveness, three follow-up emails were sent following an initial email, paper copies were mailed on request, and individual phone calls were made to follow-up. From these 800, 197 (25%) responded (114 web-based, 83 paper-based) and 127 (16%) were complete enough to use in our analyses. For the convenience sample, surveys were shared by 41 organizations, primarily over email. We received a total of 1,148 responses from the convenience sample, 672 of which included complete enough data for our analysis. In total we had 806 usable responses.
- 2020: Data were collected by both random and convenience sampling using methods similar to 2015. For the random sample, 2,000 certified organic producers were randomly selected from the the NOP database and given the option to complete the survey over email or paper. After an initial email and postcard invitation, growers received follow-up contacts by email, postcard, and phone, and hard copies of the survey sent. Survey distribution was managed by Washington State University's Social and Economic Sciences Research Center (SESRC) and distributed alongside the Organic Farming Research Foundation's (OFRF) National Organic Research Agenda. From these 2,000 surveys, 448 (22%) responded (206 web-based and 242 paper-based) and 193 (10%) were complete enough to use in our analyses. For the convenience sample, surveys were distributed by SESRC to all remaining certified organic operations over email, with postcard reminders. The survey was also advertised by OSA and OFRF. We received a total of 611 responses from the convenience sample, 215 of which included complete enough data for our analysis. In total we had 414 usable responses.

The breakdown between random and convenience sampling used in our analysis is summarized in Table 1. We consider the effect of sampling method, as well as survey distributor, through models in Section 2.

Table 1: Convenience and random samples

Year	Total complete responses	Convenience responses (%)	Random responses (%)
2010	549	549 (100 %)	0 (0 %)
2015	810	679 (84 %)	131 (16 %)
2020	413	219 (53 %)	194 (47 %)

2. Checking for biases induced by data collection methods

In order to test how our data collection methods may bias the sample, we consider the effect of the sampling approach and survey distributor in our model. First we model the effect of sampling method with our 2015 and 2020 survey data (the only survey periods where we had both sampling approaches) by including a term for responses that were gathered by random sampling compared to convenience sampling (Table 2, Model 1). We see a small and insignificant effect for sampling method, and negligible changes in the results between the models run with and without the sampling consideration (Table 2, Model 2), suggesting no significant effect on results based on sampling approach.

Regarding concerns of survey distributor (i.e. that the distribution by OSA elicited responses from a select type of organic farmer), we model this effect with our 2010 and 2015 survey data (the only survey periods that identified specific distributors) by including a term for responses that were gathered through OSA channels compared to all others (Table 2, Model 3). We see a small and insignificant effect for distributor, and negligible changes in the results between the models run with and without the distributor consideration (Table 2, Model 4), suggesting no significant effect on results based on survey distributor.

Table 2: Model results for sampling and distributor effects

	Beat regression model: Proportion of organic land planted with organic seed			
	Including sampling method (2015, 2020)	Without sampling method (2015, 2020)	Including survey distributor (2010, 2015)	Without survey distributor (2010, 2015)
	(1)	(2)	(3)	(4)
Barrier: Organic availability	−0.210***	−0.211***	−0.161***	−0.161***
	(0.042)	(0.042)	(0.041)	(0.041)
Barrier: Undesirable traits	−0.115**	−0.115**	−0.114***	−0.116***
	(0.045)	(0.045)	(0.041)	(0.041)
Barrier: Insufficient quantity	−0.049	−0.049	−0.007	−0.006
	(0.043)	(0.043)	(0.038)	(0.038)
Year: 2020 (vs. 2015)	0.108	0.113		
	(0.084)	(0.080)		
Year: 2010 (vs. 2015)			−0.225***	−0.220***
			(0.076)	(0.076)
Values organic seed	0.183***	0.182***	0.195***	0.193***
	(0.040)	(0.040)	(0.036)	(0.036)
Farm size (log acres)	−0.250***	−0.250***	−0.250***	−0.246***
	(0.054)	(0.054)	(0.052)	(0.052)
Crop diversity	0.045	0.043	0.071*	0.072*
	(0.042)	(0.041)	(0.038)	(0.038)
Certifiers request organic seed use	−0.033	−0.033	−0.119*	−0.118
	(0.076)	(0.076)	(0.072)	(0.072)
Seed saved or traded (%)	0.051	0.050	0.089***	0.088***
	(0.038)	(0.038)	(0.034)	(0.034)
Forage crops (vs. Field)	−0.985***	−0.985***	−1.023***	−1.024***
	(0.114)	(0.114)	(0.104)	(0.104)
Vegetable crops (vs. Field)	−0.579***	−0.580***	−0.571***	−0.576***
	(0.123)	(0.122)	(0.117)	(0.117)
Western region (vs. North Central)	−0.251**	−0.250**	−0.333***	−0.365***
	(0.103)	(0.102)	(0.101)	(0.097)
Northeastern region (vs. North Central)	−0.296***	−0.294***	−0.199*	−0.238**
	(0.105)	(0.104)	(0.117)	(0.112)
Southern region (vs. North Central)	−0.043	−0.044	−0.151	−0.172
	(0.136)	(0.136)	(0.130)	(0.129)
Random data collection (vs. convenience)	0.018			
	(0.090)			
Distributor: OSA (vs. Other)			−0.091	
			(0.080)	
Constant	1.033***	1.036***	1.142***	1.123***
	(0.088)	(0.087)	(0.086)	(0.084)
Observations	1,223	1,223	1,359	1,359
R^2	0.249	0.249	0.248	0.247
Log Likelihood	1,750.985	1,750.965	1,672.595	1,671.951

Note: *p<0.1; **p<0.05; ***p<0.01

3. Sample representativeness

While the mix of sampling methods creates a challenge for determining the sample's representativeness of the population, we aim to demonstrate that our sample is representative by comparing the geographic distribution of operations and farm size distribution between our sample and USDA Census data.

We first use USDA Census data from 2011, 2016, and 2021 to align with our 2010, 2015, and 2020 survey periods. Across the three survey periods, the number of respondents represented approximately 6.3%, 5.9%, and 2.5% of the total number of organic farmers identified in the USDA Census in the related years. In Table 3 we compare the number of operations in our sample (n) to the population (N) by year and region. For each census period we calculated the standard deviation of the percentage of operations represented, which we used to identify regions that are over or under-represented. Over-represented year-region combinations are signaled by green text, while under-represented year-region combinations are signaled by red text. Only in 2011 do we see variation in representation across regions, with the North Central and Western regions over-represented and the Northeast under-represented.

Table 3: Sample representativeness compared to USDA Censuses (2011, 2016, 2021)

Year	Region	Operations in sample (n)	Operations in USDA census (N)	Percent of operations represented (n/N)
2010	North Central	307	2986	10.3
2010	Northeast	51	1897	2.7
2010	South	52	540	9.6
2010	West	139	3296	4.2
2015	North Central	401	4689	8.6
2015	Northeast	142	3246	4.4
2015	South	76	1096	6.9
2015	West	191	4529	4.2
2020	North Central	178	5824	3.1
2020	Northeast	98	3888	2.5
2020	South	39	1209	3.2
2020	West	98	5273	1.9

Next, to understand farm size representativeness, we look at USDA Census organic tabulations, reported only for the 2012 and 2017 censuses.

In Figure 1 we compare these USDA Census data against our survey sample. These comparisons show a consistent over-representation of the smallest farms (1-9 acres) by 12-18% and under-representation of the largest farms. Still, our sample captures a distribution of organic farm sizes that includes medium and large farms, allowing us to understand the effect of farm size in our analysis. Additionally, an important consideration is that we calculate farm size as reported acreage planted to three major crop categories, but this approach likely underestimates acreage generally compared to USDA Census data, which reports all certified land.

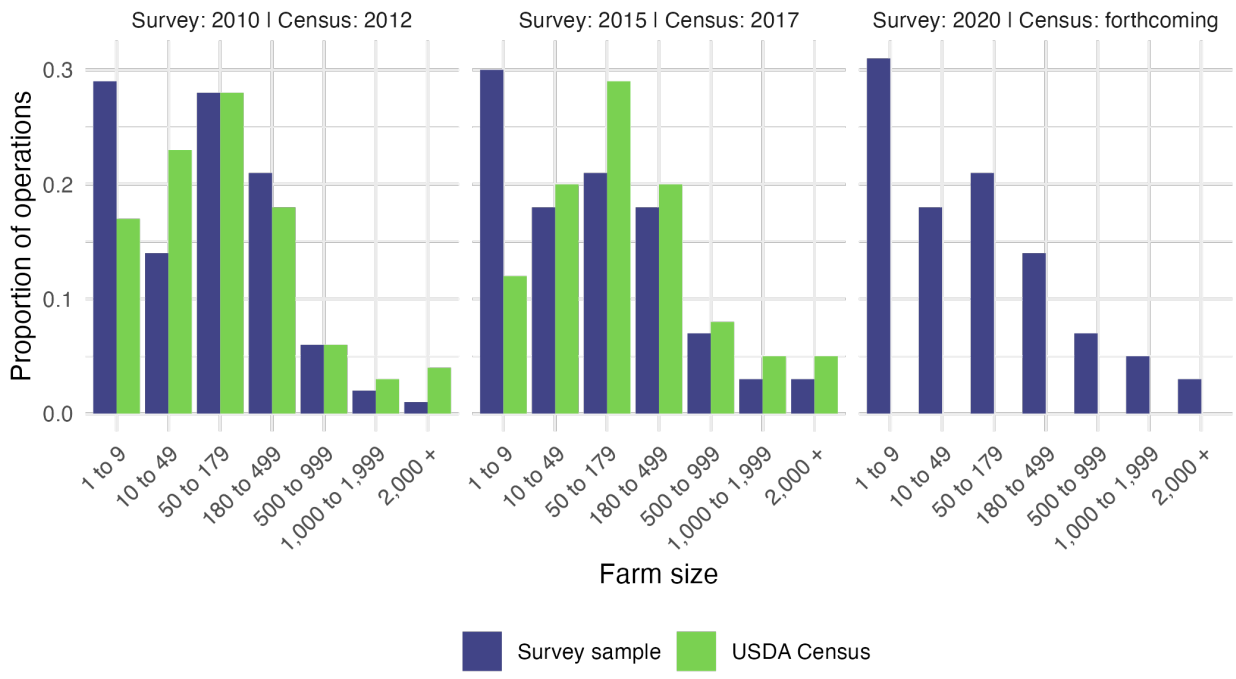


Figure 1: Sample farm size distribution compared to USDA Census data

Model checks

4. Full model results

Table 4: Final model results

	Proportion of production acreage planted with organic seed
Barrier: Organic availability	-0.193^{***}
	(0.036)
Barrier: Undesirable traits	-0.117^{***}
	(0.036)
Barrier: Insufficient quantity	-0.018
	(0.034)
Year: 2010 (vs. 2015)	-0.207^{***}
	(0.075)
Year: 2020 (vs. 2015)	0.138^{*}
	(0.079)
Values organic seed	0.176^{***}
	(0.032)
Farm size (log acres)	-0.106^{***}
	(0.020)
Crop diversity	0.077^{**}
	(0.033)
Certifiers request organic seed use	-0.079
	(0.063)
Seed saved or traded (%)	0.095^{***}
	(0.031)
Forage crops (vs. Field)	-0.904^{***}
	(0.092)
Vegetable crops (vs. Field)	-0.493^{***}
	(0.102)
Western region (vs. North Central)	-0.363^{***}
	(0.085)
Northeastern region (vs. North Central)	-0.273^{***}
	(0.092)
Southern region (vs. North Central)	-0.145
	(0.113)
Constant	1.038^{***}
	(0.077)
Observations	1,772
R^2	0.234
Log Likelihood	2,281.689
Note:	*p<0.1; **p<0.05; ***p<0.01

Model results over time

Table 5: Model results by year

	Proportion of production acreage planted with organic seed		
	2010	2015	2020
Barrier: Organic availability	−0.082	−0.191***	−0.285***
	(0.062)	(0.054)	(0.077)
Barrier: Undesirable traits	−0.133**	−0.112**	−0.122
	(0.061)	(0.053)	(0.076)
Barrier: Insufficient quantity	0.040	−0.042	−0.050
	(0.058)	(0.051)	(0.073)
Values organic seed	0.186***	0.208***	0.110
	(0.057)	(0.047)	(0.069)
Farm size (log acres)	−0.101**	−0.122***	−0.102**
	(0.039)	(0.029)	(0.045)
Crop diversity	0.150**	0.023	0.152*
	(0.060)	(0.059)	(0.086)
Certifiers request organic seed use	−0.205*	−0.065	0.048
	(0.115)	(0.092)	(0.135)
Seed saved or traded (%)	0.189***	0.027	0.102
	(0.057)	(0.048)	(0.068)
Forage crops (vs. Field)	−0.803***	−1.233***	−0.514**
	(0.158)	(0.139)	(0.203)
Vegetable crops (vs. Field)	−0.383**	−0.777***	−0.295
	(0.186)	(0.151)	(0.214)
Western region (vs. North Central)	−0.610***	−0.184	−0.345**
	(0.154)	(0.126)	(0.176)
Northeastern region (vs. North Central)	−0.092	−0.255*	−0.300*
	(0.208)	(0.134)	(0.168)
Southern region (vs. North Central)	−0.398**	−0.015	−0.089
	(0.202)	(0.167)	(0.235)
Constant	0.836***	1.247***	0.874***
	(0.123)	(0.102)	(0.147)
Observations	549	810	413
R^2	0.187	0.288	0.206
Log Likelihood	540.162	1,142.739	615.961
Note:			*p<0.1; **p<0.05; ***p<0.01

5. Considering other barriers

Table 6: Model results including additional barriers

	Proportion of production acreage planted with organic seed
Barrier: Organic availability	−0.169***
	(0.037)
Barrier: Undesirable traits	−0.108***
	(0.037)
Barrier: Insufficient quantity	0.001
	(0.035)
Barrier: Price	−0.058*
	(0.033)
Barrier: Buyer requirement	−0.114***
	(0.033)
Year: 2010 (vs. 2015)	−0.193**
	(0.076)
Year: 2020 (vs. 2015)	0.160**
	(0.081)
Values organic seed	0.160***
	(0.033)
Farm size (log acres)	−0.097***
	(0.021)
Crop diversity	0.074**
	(0.034)
Certifiers request organic seed use	−0.044
	(0.065)
Seed saved or traded (%)	0.098***
	(0.032)
Forage crops (vs. Field)	−0.904***
	(0.093)
Vegetable crops (vs. Field)	−0.469***
	(0.103)
Western region (vs. North Central)	−0.371***
	(0.086)
Northeastern region (vs. North Central)	−0.286***
	(0.094)
Southern region (vs. North Central)	−0.162
	(0.115)
Constant	1.003***
	(0.078)
Observations	1,731
R^2	0.240
Log Likelihood	2,245.812
Note:	*p<0.1; **p<0.05; ***p<0.01

6. Mediation analyses

To check for mediated effects of values and barriers on operational attributes, we run three mediation analyses. A mediating variable is one that transmits the effect one variable to another (MacKinnon et al. 2007). We test for mediation, rather than moderation through interaction effects, because we have no theoretical reason to believe that values, barriers, and operational attributes relate in a multiplicative sense.

We first addresses the potential mediation effect of larger farms on valuing organic seed. In other words, are those who value organic seed more likely to have smaller farms, and if so, how might that affect the model results? We depict a simplified representation of this mediating effect in Figure 2.

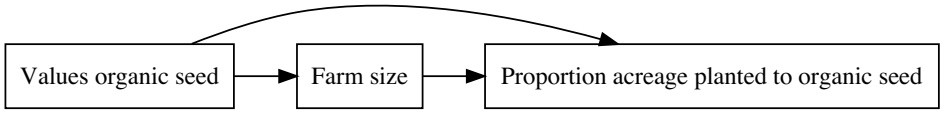


Figure 2: Mediation diagram: Organic seed value and farm size

We run a mediation analysis by comparing results of our full regression model (e.g. results in Table 5) with results of a model that regresses the same variables on the potential mediator: farm size. Results are presented in Table 6, where the Average Causal Mediation Effect (ACME) is significant. This result suggests the effect of organic seed values on organic seed use (Total Effect) is mediated by farm size.

Table 7: Mediation analysis of farm size on quantity barrier

Effects	Estimate	p-value
ACME (avg)	0.006	< 0.001
ADE (avg)	0.045	< 0.001
Prop. Mediated (avg)	0.110	< 0.001
Totel Effect	0.051	< 0.001

Next we addresses the potential mediation effect of experiencing organic seed quantity barriers on larger farms. In other words, might larger farms be more likely to experience the barrier of limited organic seed quantity, and if so, how might that affect the model results? We depict a simplified representation of this mediating effect in Figure 2.

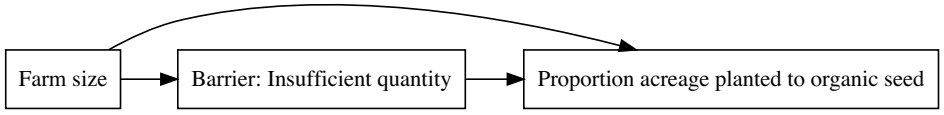


Figure 3: Mediation diagram: Farm size and organic seed quantity

We run a mediation analysis by comparing results of our full regression model (e.g. results in Table 5) with results of a model that regresses the same variables on the potential mediator: the quantity barrier. Results are presented in Table 6, where the Average Causal Mediation Effect (ACME) is small (-0.001) and not significant. The value suggests how much the effect of farm size on organic seed use (Total Effect) is mediated by this barrier. In this case, the mediated effect is small and the total and Average Direct Effects (ADE) are larger and significant. In short, the quantity barrier is not influencing the effect of farm size.

Next we addresses the potential mediation effect of experiencing organic seed availability barriers on more diverse farms. In other words, are farms that use a wider variety of seeds more likely to experience the barrier of limited organic seed availability, and if so, how might that affect the model results? We depict a simplified representation of this mediating effect in Figure 3.

Table 8: Mediation analysis of farm size on quantity barrier

Effects	Estimate	p-value
ACME (avg)	-0.001	0.58
ADE (avg)	-0.082	< 0.001
Prop. Mediated (avg)	0.016	0.58
Totel Effect	-0.084	< 0.001

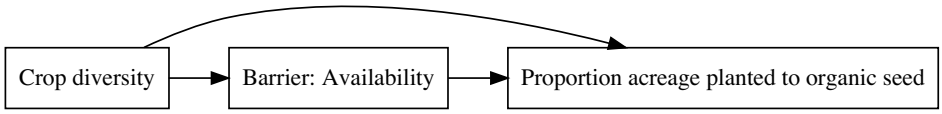


Figure 4: Mediation diagram: Crop diversity and organic seed availability

Again, we run a mediation analysis by comparing results of our full regression model with results of a model that regresses the same variables on the potential mediator: the availability barrier. Results are presented in Table 7, where again the ACME is small (-0.002) and not significant. In short, the availability barrier is not influencing the effect of crop diversity.

Table 9: Mediation analysis of crop diversity on availability barrier

Effects	Estimate	p-value
ACME (avg)	-0.002	0.096
ADE (avg)	0.027	0.014
Prop. Mediated (avg)	-0.094	0.114
Totel Effect	0.025	0.022

7. Predicted organic seed use figures

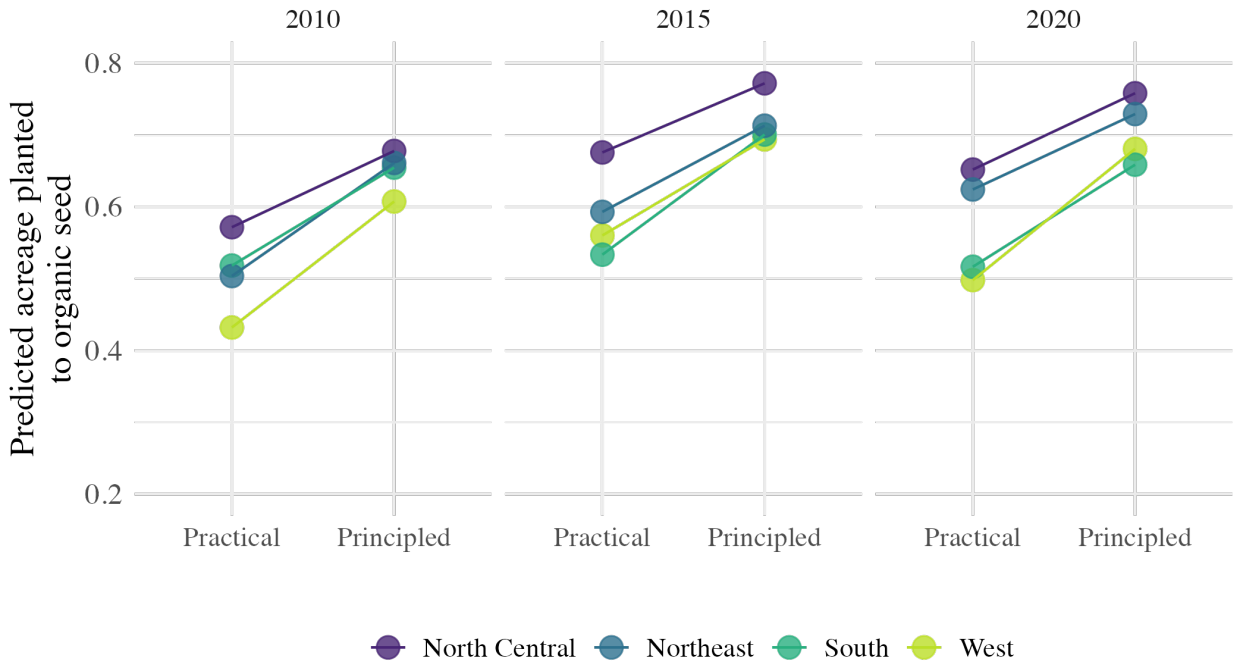


Figure 5: Predicted conventional seed use for field crop growers of two farmer profiles.

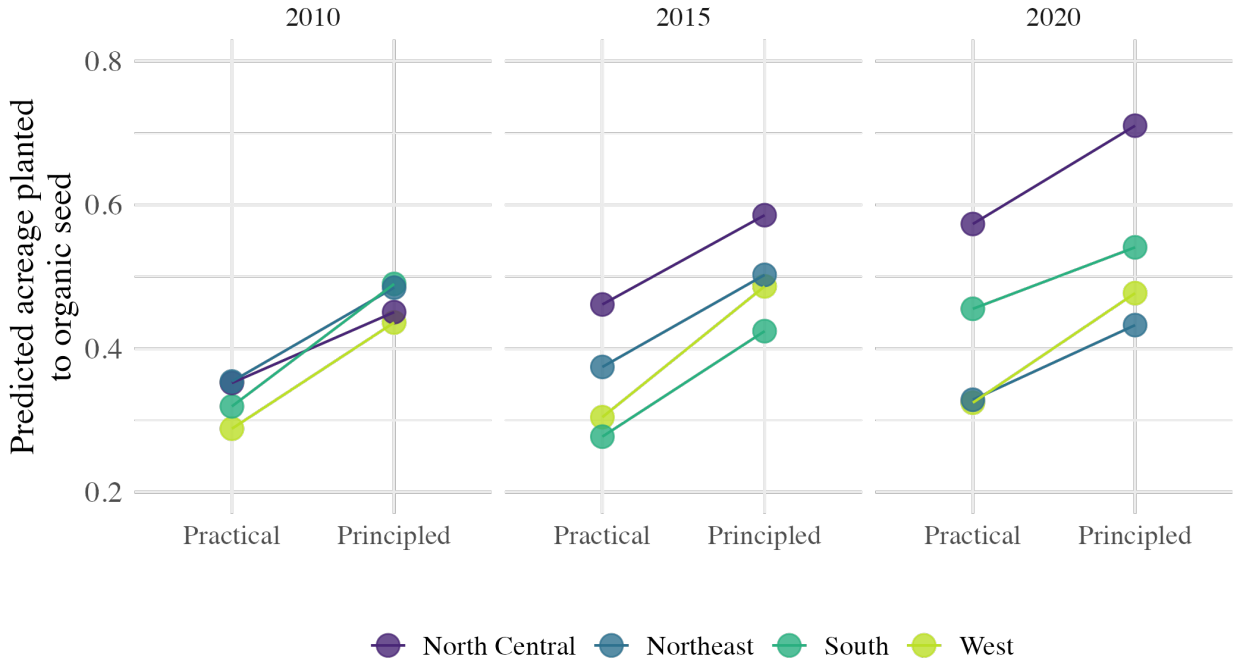


Figure 6: Predicted conventional seed use for forage crop growers of two farmer profiles.

8. Self-reported behavior change

Table 10: Self-reported increase model (versus No increase)

	Yes: Increased the %	Already at 100%
Barrier: Organic availability	0.097	-0.556^{***}
	(0.072)	(0.089)
Barrier: Undesirable traits	0.084	-0.327^{***}
	(0.065)	(0.109)
Barrier: Insufficient quantity	0.083	0.215^{**}
	(0.064)	(0.097)
Year: 2010 (vs. 2015)	1.209^{***}	-4.522^{***}
	(0.142)	(1.014)
Year: 2020 (vs. 2015)	-0.374^{**}	-0.323^{*}
	(0.149)	(0.189)
Values organic seed	0.320^{***}	0.427^{***}
	(0.060)	(0.101)
Farm size (log acres)	0.034	-0.217^{***}
	(0.039)	(0.058)
Crop diversity	−0.018	−0.034
	(0.063)	(0.092)
Certifiers request organic seed use	0.650^{***}	0.117
	(0.117)	(0.176)
Seed saved or traded (%)	-0.156^{**}	0.046
	(0.061)	(0.078)
Forage crops (vs. Field)	0.189	-0.688^{***}
	(0.172)	(0.243)
Vegetable crops (vs. Field)	0.789^{***}	-1.370^{***}
	(0.195)	(0.294)
Western region (vs. North Central)	−0.152	−0.152
	(0.157)	(0.237)
Northeastern region (vs. North Central)	−0.098	−0.358
	(0.172)	(0.248)
Southern region (vs. North Central)	−0.200	0.304
	(0.211)	(0.304)
Constant	-0.645^{***}	−0.214
	(0.149)	(0.185)

Note: *p<0.1; **p<0.05; ***p<0.01

9. Certifiers' role

An additional consideration in our analysis is the effect of certifier. We expect certifiers to enforce the regulation differently **(cite)**, but we do not have data that allows us to specifically identify each producers' certifier in our data for use in our main analysis. However, we do have some suggestion of growers' certifier based on our data collection process in certain survey periods. When data are available, we assume that respondents who completed the survey distributed by a certain certifier are certified by them. In Table 9, Model 1 we model the effect of five distinct certifiers and one 'other' category, compared against compared to a baseline of the most popular certifier in our limited dataset. We see a small and insignificant effect for different certifiers compared to a baseline certifier group, and negligible changes in the results between the models run with and without the sampling consideration (Table 9, Model 2).

Table 11: Model checks for certifier effects

	Proportion of production acreage planted with organic seed	
	(1)	(2)
Barrier: Organic availability	−0.142**	−0.146**
	(0.068)	(0.068)
Barrier: Undesirable traits	−0.051	−0.051
	(0.068)	(0.068)
Barrier: Insufficient quantity	−0.021	−0.022
	(0.064)	(0.063)
Year: 2010 (vs. 2015)	−0.232	−0.278**
	(0.224)	(0.136)
Year: 2020 (vs. 2015)	−0.005	0.0004
	(0.287)	(0.247)
Values organic seed	0.169***	0.167***
	(0.061)	(0.061)
Farm size (log acres)	−0.153***	−0.156***
	(0.043)	(0.043)
Crop diversity	0.153**	0.154**
	(0.064)	(0.063)
Seed saved or traded (%)	0.067	0.059
	(0.063)	(0.059)
Forage crops (vs. Field)	−0.431***	−0.420***
	(0.121)	(0.120)
Vegetable crops (vs. Field)	−0.872***	−0.861***
	(0.160)	(0.158)
Western region (vs. North Central)	−0.699***	−0.675***
	(0.204)	(0.197)
Northeastern region (vs. North Central)	−0.417	−0.357**
	(0.271)	(0.179)
Southern region (vs. North Central)	−0.354	−0.283
	(0.247)	(0.215)
Certifier Other	−0.319	−0.023
	(0.395)	(0.290)
Certifier 24	0.094	
	(0.308)	
Certifier 2	0.436	
	(0.446)	
Certifier 27	0.159	
	(0.310)	
Certifier 42	0.021	
	(0.256)	
Certifier 7	−0.152	
	(0.246)	
Constant	−0.002	
	(0.344)	
Constant	1.335***	1.350***
	(0.238)	(0.150)
Observations	491	491
R^2	0.266	0.262
Log Likelihood	640.502	639.538

Note: *p<0.1; **p<0.05; ***p<0.01

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